# Nathalie C. Haurberg

Associate Professor of Astronomy · Department of Physics and Astronomy, Knox College, Galesburg, IL
nhaurber@knox.edu · www.nathaliehaurberg.com · ORCID 0000-0001-5247-1371

## Education

**Ph.D.**, Astronomy (second concentration in Physics), Indiana University, Bloomington, IN 2013
*Chemical Abundances and Star-Formation Histories of Low-Mass Dwarf Galaxies*; advisor: Dr. John Salzer
**M.A.**, Astronomy, Indiana University, Bloomington, IN 2008
**B.A.**, Physics (second concentration in Comparative Politics), *cum laude*, Knox College, Galesburg, IL 2006

## Appointments and Leadership

**Associate Professor of Astronomy**, Knox College *(title in Physics, 2019–2025)* 2019 – Present
**Assistant Professor of Physics**, Knox College 2014 – 2019
**Visiting Instructor / Visiting Professor of Physics**, Knox College 2013 – 2014
**Director, MACRO Consortium** *(Board of Directors, 2025–2026; Faculty Lead, 2024–2025)* 2026 – Present
Five-institution consortium operating the 0.5-m Robert L. Mutel Telescope (Sonoita, AZ) for collaborative undergraduate research and teaching.
**Director, Knox College Observatory**; Program Advisor, Astronomy
**Academic Director**, Summer Science Program in Astrophysics *(Associate Academic Director, 2025)* 2026

## Refereed Publications

(My name is **bolded**; long author lists are shortened. "MACRO Consortium" marks papers published under the collective consortium author, of which I am a member.)

1. Golay, W. W., Cannon, J. M., Mutel, R. L., **Haurberg, N. C.**, et al., and the MACRO Consortium 2026, "Phase-Recurrent Coherent Radio Pulses From Polars." *Nature Astronomy*, under review.
2. Groene, I. J., et al., and the MACRO Consortium 2026, "The Soft X-ray Properties of AR Ursae Majoris in a High State." *ApJ*, under review.
3. Shepley, M., Gormanous, D. G., Salzer, J. J., **Haurberg, N. C.**, et al. 2026, "H$\alpha$ Star Formation Rates of the SHIELD Dwarf Galaxies." *ApJ*, in press.
4. Gunderson, S. J., et al., and the MACRO Consortium 2026, "A Multiwavelength View of $\rho$ Oph II: Disentangling the Variability of the Multi-Star Component C." *ApJ*, 1011, 108. doi:10.3847/1538-4357/aeac15
5. Gormanous, D. G., Dziarski, A. S., Salzer, J. J., Cannon, J. M., Janowiecki, S., **Haurberg, N. C.**, et al. 2026, "UV Star Formation Rates of the SHIELD Dwarf Galaxies." *ApJ*, 1006, 90. doi:10.3847/1538-4357/ae8092
6. Gunderson, S. J., Codd, J., Golay, W. W., [. . . ], **Haurberg, N. C.**, et al., and the MACRO Consortium 2025, "A Multiwavelength View of $\rho$ Oph I: Resolving the X-ray Source Between A and B." *ApJ*, 995, 13. doi:10.3847/1538-4357/ae0d7e
7. Hirschauer, A. S., Salzer, J. J., **Haurberg, N. C.**, Gronwall, C., Janowiecki, S. 2022, "H$\alpha$ Dots: Direct-method Metal Abundances of Low-luminosity Star-forming Systems." *ApJ*, 925, 131. doi:10.3847/1538-4357/ac402a
8. McNichols, A. T., Teich, Y. G., Nims, E., [. . . ], **Haurberg, N. C.**, et al. 2016, "SHIELD: Neutral Gas Kinematics and Dynamics." *ApJ*, 832, 89. doi:10.3847/0004-637X/832/1/89
9. Teich, Y. G., McNichols, A. T., Nims, E., [. . . ], **Haurberg, N. C.**, et al. 2016, "SHIELD: Comparing Gas and Star Formation in Low-mass Galaxies." *ApJ*, 832, 85. doi:10.3847/0004-637X/832/1/85
10. **Haurberg, N. C.**, Salzer, J. J., Cannon, J. M., Marshall, M. V. 2015, "Oxygen Abundance Measurements of SHIELD Galaxies." *ApJ*, 800, 121. doi:10.1088/0004-637X/800/2/121
11. Skillman, E. D., Salzer, J. J., Berg, D. A., Pogge, R. W., **Haurberg, N. C.**, et al. 2013, "ALFALFA Discovery of the Nearby Gas-rich Dwarf Galaxy Leo P. III. An Extremely Metal Deficient Galaxy." *AJ*, 146, 3. doi:10.1088/0004-6256/146/1/3
12. Rhode, K. L., Salzer, J. J., **Haurberg, N. C.**, et al. 2013, "ALFALFA Discovery of the Nearby Gas-rich Dwarf Galaxy Leo P. II: Optical Observations." *AJ*, 145, 149. doi:10.1088/0004-6256/145/6/149
13. **Haurberg, N. C.**, Rosenberg, J., Salzer, J. J. 2013, "Metal Abundances of 12 Dwarf Irregulars from the ADBS Survey." *ApJ*, 765, 66. doi:10.1088/0004-637X/765/1/66
14. **Haurberg, N. C.**, Lubell, G. M. G., Cohn, H. N., Lugger, P. M., et al. 2010, "Ultraviolet-bright Stellar Populations and Their Evolutionary Implications in the Collapsed-core Cluster M15." *ApJ*, 722, 158. doi:10.1088/0004-637X/722/1/158

## Recent Observing Time

- **PI**, "Fanning the Flames – A Radio Survey of Pre-Polars," VLA/27A-348 (40.4 hr); and Observing for University Classes, VLA/TDEM0044 (2 hr).
- **Co-I** on eight further programs, 2025–2027, on the VLA, VLBA, and MMT (about 270 hr in total), studying radio pulses and accretion states in polars and intermediate polars, and radio emission from massive and Be stars.

## Selected Presentations

- "Seeing Double: Untangling Accreting Binary Stars in Optical and Radio," invited talk, Peoria Astronomical Society, Nov. 2026.
- "Supporting Student Research Through Simultaneous Optical & Radio Observations," *Small Campus Observatories as Promoters of Education in Science (SCOPES)*, June 2025.
- "Simultaneous Optical and Radio Observations of Magnetic Cataclysmic Variables by the MACRO Consortium," Joint Meeting of the Iowa and Illinois Sections of the AAPT, April 2025.
- "The Diversity of Research at MACRO: Science Highlights," 244$^{\text{th}}$ Meeting of the American Astronomical Society, June 2024.
- "Finding Galaxies with Hydrogen Gas," 67$^{\text{th}}$ Annual William Thomas Jackson Lecture in Science, Eureka College, April 2021.
- Fifteen conference abstracts and posters (2010–2027), many with student co-authors, and eleven invited colloquia in total.

## Grants, Fellowships, and Honors

| | |
|---|---|
| Ellen Browning Scripps Foundation, "Compact Grism Spectroscopy: Bringing Light into the Classroom" | 2026 |
| Phi Beta Kappa, Alumni Inductee | 2025 |
| Knox College Exceptional Achievement Award | 2024 |
| G. L. Vitale Family Foundation and Ellen Browning Scripps Foundation, Knox observatory and dome (co-recipient) | 2019 |
| Knox College Cartier Award | 2019 |
| Knox Faculty Research Grant; Hummel Foundation Grant; Billy W. Geer Research Fund Award | 2013 – 2018 |
| Indiana University Dissertation Year Fellowship; Indiana Space Grant Consortium Graduate Fellowships | 2009 – 2013 |

## Teaching and Mentoring

Courses at Knox College: Astronomy; The Search for Extraterrestrial Life; Observational Astronomy; Stellar Astrophysics; Extragalactic Astrophysics; Introduction to Research; Mechanics; Heat, Waves, and Light; Electricity & Magnetism; Modern Physics; Optics; Thermodynamics & Statistical Mechanics; Classical Dynamics; seminars in Analytical Mechanics and Quantum Mechanics; Science-Fiction and Human Identity *(interdisciplinary)*.

Mentoring: Honors projects and independent research with undergraduates in physics, astronomy, and computer science, leading to co-authored refereed papers and AAS posters. MACRO Summer Program (2024–2026; 7–16 students per year) on optical and radio variability of magnetic cataclysmic variables; inter-institutional research course on Be stars (2025); Knox Summer Collaborative Research Projects (2021, 2023).

## Service and Outreach

- **Professional:** outside member of a Ph.D. committee, Department of Physics & Astronomy, University of Iowa (2026–present); referee, *American Journal of Undergraduate Research* (2025); external tenure and promotion review, Washington & Jefferson College, and external program review, Department of Physics, Principia College (2021).
- **Knox College:** Chair, Campus Diversity Committee (2019–2020, 2024–2025); Chair, Student Life Committee (2018); AI Task Force core committee (2026); Faculty Experience Committee (2025–present); five search committees, including the Presidential and CFO searches; faculty advisor, Physics and Astronomy Club.
- **Outreach:** public viewing nights at the Knox College Observatory four to five times a year since 2020, with attendance regularly near 100; public talks for community groups and libraries; NASA International Year of Astronomy Ambassador (2009).
- **Memberships:** American Astronomical Society; American Association of Physics Teachers.